

When Not to Reconfigure: Sequential Expert Routing for Dynamic Warehouse Slotting under Non-Stationary Demand

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Abstract

Dynamic slotting in fast-moving consumer goods (FMCG) warehouses is typically optimized per-period or over short rolling horizons, treating each re-slotting decision as independent of its predecessors. We show that this myopic approach can be globally suboptimal when physical reconfiguration costs create inter-temporal coupling: a locally beneficial reconfiguration made before a demand or cost shock can lock the warehouse into expensive future adjustments. We formalize this as the *Dynamic Warehouse Expert Routing Problem* (DW-ERP), a warehouse-specific instance of switching-cost online optimization in which a library of operations research (OR) experts proposes slotting actions each period, and the decision is which expert to follow. Through controlled experiments validated against exact enumeration on small instances, we characterize the conditions under which myopic expert selection fails: inter-temporal traps concentrate at intermediate lead times before cost shocks (1.26–1.69% normalized trap gain), the full-information opportunity is modest (~ 1 –5%) but reproducible, and traps are entirely absent without cost shocks. Counter-intuitively, deployable receding-horizon policies with crude internal cost models capture most of this opportunity not by forecasting better, but by being conservative—model imprecision acts as implicit stickiness that avoids over-reconfiguration. We empirically demonstrate that algorithm-switch counts are an inadequate surrogate for physical reconfiguration cost (hidden reconfiguration rate 17–69%), and that value-per-move, not move frequency, is the correct selective-reconfiguration metric. On real warehouse data (WEPA, 3 months, 411,830 orders), no inter-temporal trap manifests under natural conditions, bounding trap-aware expert routing to genuinely non-stationary scenarios such as promotion seasons and product-line transitions.

1 Introduction

Warehouse slotting—the assignment of stock-keeping units (SKUs) to storage locations—is a classical operations research problem whose solution directly determines order-picking travel distance, replenishment frequency, and congestion. In FMCG distribution centers, demand is non-stationary (promotions, seasonal shifts, product lifecycle turnover), so the optimal layout changes over time. *Dynamic slotting* methods address this by re-optimizing the layout periodically or over rolling horizons.

The prevailing approach treats each re-optimization as an independent instance: given the current demand forecast, solve for the best layout. This is *myopic* in the technical sense—it minimizes the current period’s cost without considering how today’s reconfiguration changes tomorrow’s options. When physical reconfiguration is free, myopia is harmless. When it is costly, a tension emerges: a reconfiguration that is locally beneficial (reduces current-period picking cost) may be globally harmful if it must be partially undone at high cost when demand shifts again.

We call this phenomenon an *inter-temporal reconfiguration trap* (Section 7). The core question of this paper is:

When should a warehouse defer a locally optimal physical reconfiguration, and how much of the resulting full-information opportunity can a

deployable policy capture?

This question sits at the intersection of three literatures: metrical task systems and smoothed online optimization (switching costs), warehouse online reoptimization, and algorithm selection (Section 2). Our contribution is not a new sequential decision theory—MTS/SOCO provides that—but a warehouse-specific instantiation with discrete physical layout states, heterogeneous OR experts as candidate policies, real reconfiguration costs, non-stationary demand, and a detailed empirical characterization of when myopic optimization fails and when it does not.

1.1 Contributions

1. **Formulation.** We define DWERP as $\pi(S_t) \rightarrow E_t \rightarrow A_t$, $S_{t+1} = f(S_t, A_t, \xi_{t+1})$, with transition cost $C_{\text{trans}} = d(L_t, L_{t+1})$ measured in physical layout distance, not algorithm-switch indicators (Section 3).
2. **Trap characterization.** Through controlled experiments validated against exhaustive enumeration (beam search = exact optimum on small instances, gap 0.00%), we show: (i) traps exist (constructive evidence: accepting a 1.6% current-period loss avoids 50 costly moves, yielding 5.34% long-term gain); (ii) they are rare-but-costly (1/12 seeds material, insurance-shaped distribution); (iii) they concentrate at intermediate lead times ($\Delta t = 1$ day between demand transition and cost shock); (iv) they vanish without cost shocks (Sections 7, 7.4).
3. **The deployment paradox.** Deployable receding-horizon policies with crude internal cost models capture most of the full-information opportunity—not through better forecasting, but through implicit conservatism. Model fidelity does not gate capture (Section 8).
4. **Transition cost surrogate failure.** Algorithm-switch counts $\mathbb{1}[E_t \neq E_{t-1}]$ are an inadequate proxy for physical reconfiguration: hidden reconfiguration (same expert name, massively different layout) occurs in 17–69% of periods (Section 10).
5. **Real-data boundary.** On WEPA real warehouse data (411,830 orders, 3 months), no trap manifests under natural or moderately stressed conditions—bounding applicability to genuinely non-stationary events (Section 11).

2 Related Work

2.1 Metrical Task Systems and Switching-Cost Online Optimization

Metrical task systems (MTS) (Borodin et al., 1992) model an online decision-maker choosing a state each period from a metric space, paying both service costs and movement costs. Smoothed online convex optimization (SOCO) (Goel and Wierman, 2009) extends this with continuous action spaces. Learning-augmented MTS uses predictions to improve competitive ratios (Bubeck et al., 2021). Our DWERP is a warehouse-specific MTS with discrete layout states, but we do not claim new online-algorithm theory; rather, we instantiate MTS structure in a domain with hard capacity constraints, heterogeneous candidate policies, and physical reconfiguration costs that violate clean metric assumptions (Section 10).

2.2 Warehouse Online Reoptimization

Recent work studies online reoptimization in picking operations (Lorenz et al., 2024a): simple re-solving of batching and routing problems as orders arrive dynamically performs surprisingly well. Related work on anticipation (Lorenz et al., 2024b) analyzes strategic waiting by pickers and finds it can be counterproductive. Our *reconfiguration deferral* is fundamentally different from *picker*

waiting: we defer physical warehouse state changes (pallet moves, pick-face reassignments), not labor scheduling decisions. The two share the inter-temporal structure but have different cost models and decision variables.

2.3 Dynamic Slotting and Algorithm Selection

Dynamic storage location assignment has evolved from turnover-based ABC (Heskett, 1963) through affinity-based methods (Kofler et al., 2014; Mirzaei et al., 2022) to DRL-based policies (Waubert de Puiseau et al., 2022). Algorithm selection—choosing which solver to use per instance—is studied in OR (?) and hyper-heuristics. Closest to our setting, contextual bandit put-away policies (Yuan et al., 2024) and future-order-driven repositioning (Karimi et al., 2026) adapt storage decisions to changing demand. None of these treat the *stateful reconfiguration trajectory* as the decision object with explicit switching costs; they optimize per-instance or per-episode independently.

3 Problem Formulation

3.1 DWERP

Let $t = 1, \dots, T$ index planning periods (days or shifts). The warehouse state is

$$S_t = (\mathcal{O}_t, D_t, F_t, I_t, L_t, R_t, \mathcal{C}) \quad (1)$$

where $\mathcal{O}_t$ = observed orders, D_t = demand history, F_t = forecast, I_t = inventory, L_t = physical layout (SKU→location assignment), R_t = resources (labor, equipment, move-cost schedule), and $\mathcal{C}$ = hard constraints (capacity, zone, FEFO).

A library $\mathcal{E} = \{E_1, \dots, E_K\}$ of OR experts is available. Each expert E_i is a *policy*: it reads the state and outputs a new layout $L'_t = E_i(S_t)$ together with a cost decomposition and feasibility check.

The routing policy selects one expert per period:

$$\pi(S_t) \rightarrow E_t, \quad A_t = E_t(S_t), \quad S_{t+1} = f(S_t, A_t, \xi_{t+1}) \quad (2)$$

where ξ_{t+1} is the next period’s demand/cost realization. The long-run objective is:

$$\min_{\pi} \sum_{t=1}^T \left[\underbrace{C_{\text{op}}(S_t, A_t)}_{\text{picking}} + \underbrace{C_{\text{trans}}(L_{t-1}, L_t)}_{\text{reconfiguration}} \right] \quad (3)$$

3.2 Transition Cost

We define $C_{\text{trans}} = d(L_{t-1}, L_t)$ as a function of physical layout distance—not the algorithm-switch indicator $\mathbb{1}[E_t \neq E_{t-1}]$. In Section 10 we empirically justify this choice. Our primary distance proxy is the number of SKU relocations (Hamming distance over the layout map):

$$d(L, L') = |\{s : L(s) \neq L'(s)\}| \quad (4)$$

with cost $C_{\text{trans}} = \lambda_m \cdot d(L_{t-1}, L_t) \cdot \text{mc}_t$, where mc_t is the per-move cost at time t (including regime-specific multipliers).

3.3 Information Regimes

- **Ex-post Beam Full-Information Policy (BFIP)**: sees the complete future trajectory $S_{1:T}$; serves as the (approximate) upper bound. Beam search with myopic-trajectory injection guarantees $C_{\text{BFIP}} \leq C_{\text{Myopic}}$.

- **Receding-Horizon Warehouse Policy (RHC)**: sees only Information_t (current state, known promotions, forecasts, known cost schedules); optimizes over a lookahead window H using an internal cost model; executes the first action. This is the deployable policy class.
- **Myopic (ex-post)**: per-period argmin over realized costs; clairvoyant but greedy. The trap victim.
- **Greedy-FC**: per-period argmin over forecast-model costs; the realistic deployable baseline.

4 Expert Library

Table 1: OR Expert Library v3.1. All experts are capacity-constrained ($\lceil n/L \rceil$ per location).

ID	Expert	Decision rule	Best regime
E1	Static ABC	rank by full-history frequency	stable, high move cost
E2	COI	rank by frequency / volume	space-constrained
E3	Affinity	recency-weighted co-pick clustering	strong baskets
E4	Forecast	rank by informed forecast p_{50}	known promotions
E5	Robust	rank by $p_{50} - \kappa \cdot (p_{90} - p_{10})$	high uncertainty
E6	DDSR-lite	opportunistic payback-gated swaps	known-order horizon
E7	Joint	CP-SAT: forecast cost + move penalty	cost-coupled

Experts E1–E5 produce layouts independent of the incoming layout (cached per period). E6 (DDSR-inspired) and E7 (joint optimizer) read the current layout: E6 swaps only when expected saving exceeds a payback threshold; E7 solves a CP-SAT model with explicit move penalties.

5 Experimental Environment

5.1 Synthetic Platform (Primary)

World: 120 SKUs (Zipf-distributed demand, category basket structure with concentration 0.7), 60 rack locations (6m bay spacing, $\sim 120\text{m}$ depth), 28-day horizon.

Regime sequence (8 phases): stable(4d) $\rightarrow$ promo ramp(2d) $\rightarrow$ promo peak(4d) $\rightarrow$ promo decay(2d) $\rightarrow$ stable(3d) $\rightarrow$ velocity reversal(6d) $\rightarrow$ affinity shift(4d) $\rightarrow$ move-cost shock(3d, $\times 20$). Promotions are *known events* (multiplied into forecasts); demand transitions ramp gradually.

Evaluation: per-period realized cost = L0 route distance (greedy TSP per order) + $\lambda_m \cdot \text{moves} \cdot \text{mc_unit}$. Beam width 30 (validated against exact enumeration, Section 7.4); deterministic CP-SAT (single worker, deterministic time budget).

Protocol discipline: experts slot on *history* only (known-at-time $\leq t$); evaluation on *future* orders; capacity audit on every plan; non-vacuous validation (empty fact tables rejected).

5.2 Real Data (Supplementary)

Instacart (Track B): 3.4M baskets, user-level train/eval split (no identity leakage), top-120 SKUs by train-side frequency. Real basket co-occurrence concentration: 0.23 (vs synthetic 0.70).

WEPA (external validity): real block-stacking warehouse layout (1,074 storage cells, 1.4m grid) + 411,830 retrieval orders over 3 months, bundled with the SLAPStack package (WEPAS-tacks, 2024).

6 Expert Diversity

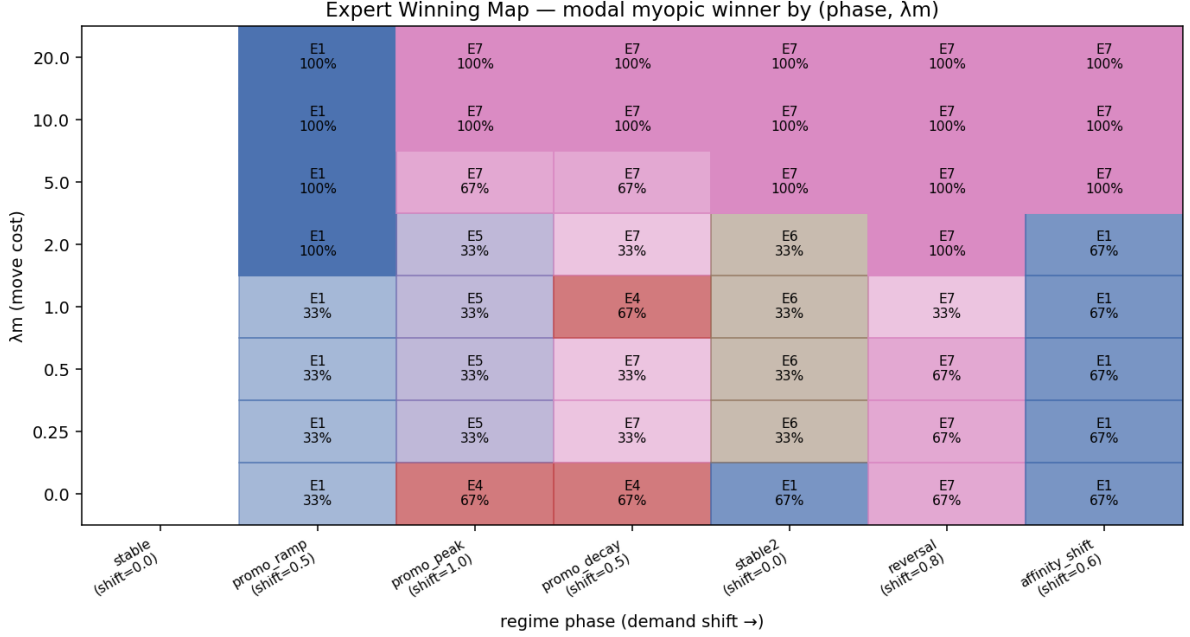


Figure 1: Expert Winning Map: modal myopic winner by regime phase (x-axis, demand shift) and move-cost multiplier λ_m (y-axis). Cell color = expert ID; opacity = stability (share of modal winner across seeds). No single expert dominates; winners switch systematically with state.

Finding 1 (T0: No Dominant Expert). *On the synthetic regime sequence (3 seeds $\times$ 7 evaluated phases), the top expert (E7 Joint) wins only 52% of periods, with 6 distinct winners observed. Winner switches align with regime phases: promotion $\rightarrow$ E3/E4 family, stable $\rightarrow$ E1, move-cost shock $\rightarrow$ E7 (100%).*

This establishes that a fixed-expert policy is suboptimal—but does not yet establish that *sequential* routing adds value over per-period selection. That is the subject of Section 7.

7 Myopic Failure: The Trap Window

7.1 Definitions

Definition 1 (Normalized Trap Gain). *For a future window H starting at period t :*

$$NTG_t(H) = \frac{C_{t:t+H}^{Myopic} - C_{t:t+H}^{Dynamic}}{C_{t:t+H}^{Myopic}} \quad (5)$$

Definition 2 (Material Trap). *MaterialTrap $_t = \mathbb{1}[NTG_t > \tau]$ with $\tau = 1\%$.*

7.2 Existence (T1a)

Finding 2 (Constructive Evidence). *On seed 17, the dynamic (BFIP) trajectory deliberately accepts a 1.6% current-period loss in the affinity-shift phase (choosing E1 over E3), which positions the layout so that the subsequent move-cost-shock period requires only 5 moves instead of 55. Total long-term gain: 5.34% (TrapScore = 2116/80 = 26.3).*

This is the core mechanism in miniature: *myopic optimization over-configures to current demand, then pays dearly to undo it when costs spike.*

7.3 Prevalence (T1b)

Finding 3 (Insurance-Shaped Distribution). *Across 12 independent seeds: 10/12 have gap ≈ 0 , 1/12 small (0.24–1.2%), 1/12 large (5.34%). Divergence points concentrate at promo-ramp (4/12) and stable-2 (4/12)—the “eve of structural shifts.” Material traps: 1/12 seeds.*

The value of sequential routing is *event-dependent*—like insurance, it costs nothing when nothing happens and saves substantially when a trap does occur. This motivates the controlled experiments of Section 7.4.

7.4 Exact Validation

On small instances (60 SKU, 30 locations, 4 evaluated periods, $7^4 = 2,401$ trajectories exhaustively enumerated):

Finding 4 (Beam = Exact). *Beam-30 achieves the exact optimum on all 4 seeds (gap 0.00%). **With** move-cost shock ($\times 20$): myopic gap 0.46–2.68%, trajectories diverge. **Without** shock: exact = beam = myopic (gap 0.00% everywhere).*

The no-shock control confirms the causal structure: *traps are purely the interaction of transition-period decisions with subsequent cost shocks*—neither alone suffices.

7.5 Controlled Trap Phase Diagram

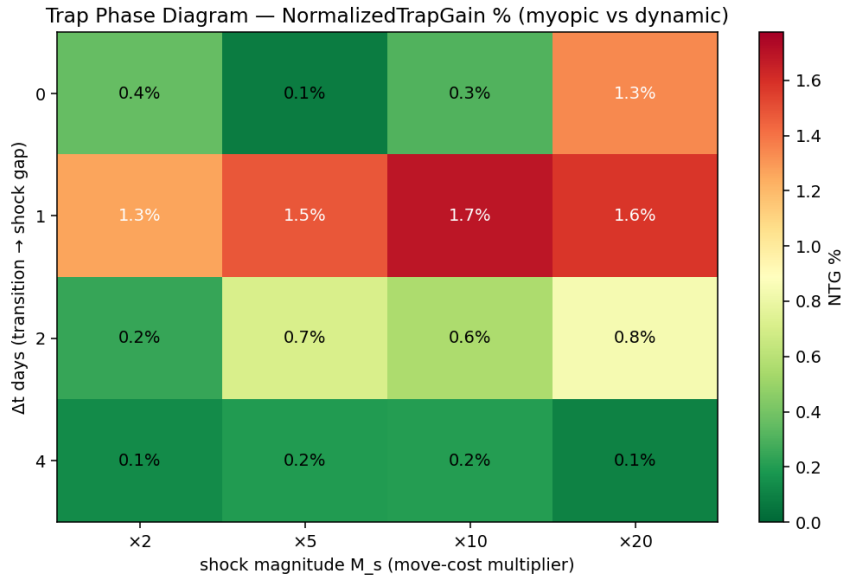


Figure 2: Trap Phase Diagram: NormalizedTrapGain (%) across lead time Δt (transition-to-shock gap, rows) and shock magnitude M_s (columns). Traps concentrate at the intermediate lead time $\Delta t = 1$ (NTG 1.26–1.69%, all magnitudes material); $\Delta t \geq 2$ decays; $\Delta t = 0$ requires extreme shock only.

Systematically varying lead time $\Delta t \in \{0, 1, 2, 4\}$ (days between demand transition and cost shock) and shock magnitude $M_s \in \{2, 5, 10, 20\}$:

Finding 5 (Intermediate Lead-Time Band). *Traps concentrate at $\Delta t = 1$: all four magnitudes are material (NTG 1.26–1.69%). At $\Delta t = 0$, only $M_s = 20$ is material (1.34%). At $\Delta t \geq 2$, no cell is material. The myopic signature: zero moves during transition, then $2\times$ dynamic’s moves at shock (67 vs 35 at $\Delta t = 1, M_s = 2$).*

The non-monotonicity in Δt is itself informative: at $\Delta t = 0$ the shock arrives before any reconfiguration can “set wrong”; at $\Delta t \geq 2$ the wrong reconfiguration has time to be amortized. The dangerous zone is *just enough lead time to act, not enough to recover*.

8 The Deployment Paradox

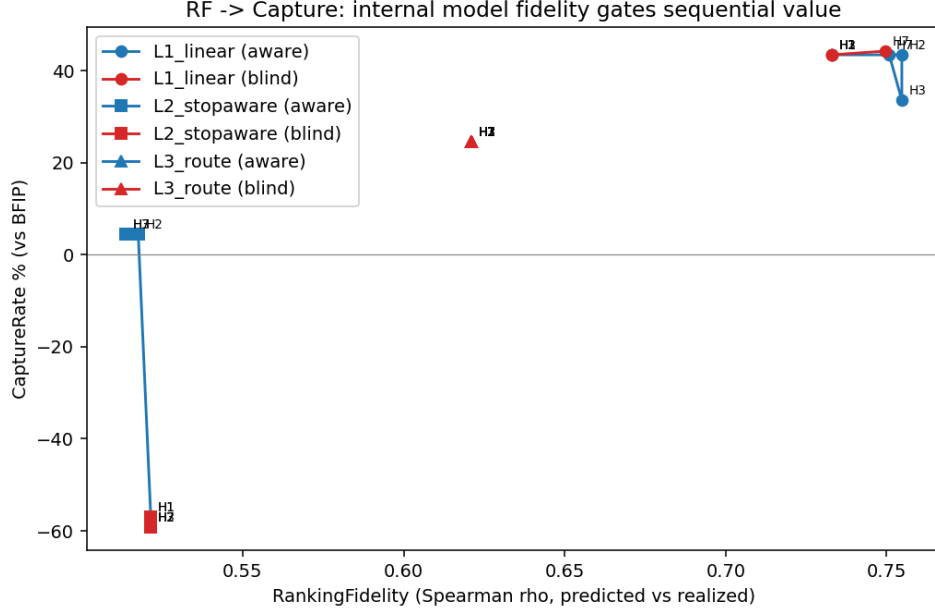


Figure 3: Ranking Fidelity vs. Capture Rate: each point is a (model, horizon) configuration; x-axis = Spearman ρ between predicted and realized per-expert cost rankings; y-axis = Oracle Opportunity Capture Rate. Capture does not increase monotonically with ρ : the crudest model (L1) achieves the highest capture via implicit conservatism.

8.1 Capture Rate

Definition 3 (Oracle Opportunity Capture Rate).

$$\text{CaptureRate} = \frac{C_{\text{GreedyFC}} - C_{\text{RHC}}}{C_{\text{GreedyFC}} - C_{\text{BFIP}}} \quad (6)$$

8.2 Results

Twelve seeds, receding-horizon $H = 2$ with three internal cost models (linear $\sum p_{50} \cdot \text{dist}$; stop-count-aware; route-aware), schedule-aware and schedule-blind:

Finding 6 (Capture by Conservative Models). *The crudest model (L1 linear) captures the most opportunity relative to ex-post myopic (mean +43.4% on seeds with headroom); the route-aware model captures +24.6%; the stop-aware model is unstable. Model fidelity does not gate capture.*

Finding 7 (The Lazy Policy Paradox). *On seed 17: ex-post myopic = 38,106, RHC = greedyFC = 36,444, BFIP = 36,070. The deployable policies are not better forecasters—they are less aggressive reconfigurers. Their imprecision acts as implicit stickiness that avoids the trap. The true clairvoyance premium (BFIP vs greedyFC) is only $\sim 1.0\%$.*

This connects directly to the MTS literature’s lazy-vs-aggressive trade-off: in switching-cost environments, *conservatism is self-protecting*. Our experiments provide the warehouse-domain empirical instance of this theoretical result.

9 Reconfiguration Sensitivity

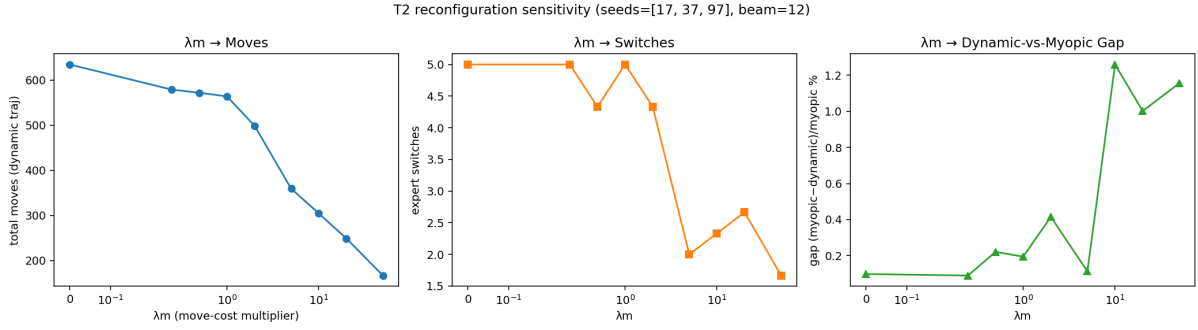


Figure 4: Reconfiguration sensitivity curves: $\lambda_m \rightarrow$ total moves (left), expert switches (center), and dynamic-vs-myopic gap (right, three statistics). Gap peaks at $\lambda_m = 10$ and plateaus at high λ_m (dynamic moves more but better).

Sweeping the global move-cost multiplier $\lambda_m \in \{0, 0.25, 0.5, 1, 2, 5, 10, 20, 50\}$ (3 seeds, sequence-internal shocks disabled):

Finding 8 (Left-Low, Mid-Peak, Right-Plateau). *The dynamic-vs-myopic gap is $\sim 0.1\%$ at $\lambda_m \leq 0.25$ (moves nearly free, planning unnecessary), peaks at $\lambda_m = 10$ (mean 1.26%, max 3.47%), and plateaus at 1.0–1.2% for $\lambda_m \in [20, 50]$. At high move costs, dynamic moves more than myopic but better: 164 vs 152 moves at $\lambda_m = 50$.*

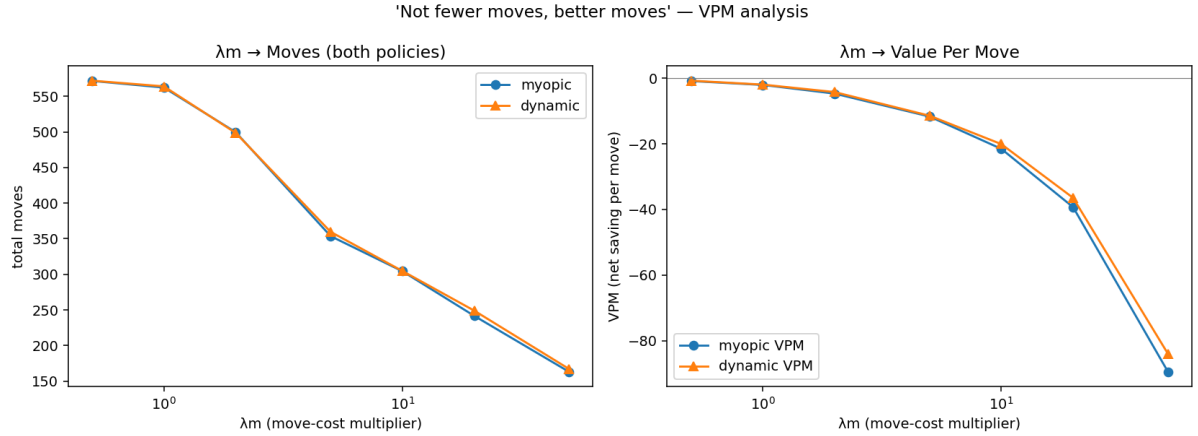


Figure 5: Value Per Move analysis: $\lambda_m \rightarrow$ total moves (left) and VPM (right) for myopic and dynamic policies. VPM is negative against the never-move baseline at all λ_m (top-costs: even optimal moves lose vs. inaction), but $VPM_{\text{dynamic}} > VPM_{\text{myopic}}$ at all levels with widening gap (+0.1 $\rightarrow$ +5.5).

Finding 9 (Value Per Move). $VPM = (\Delta C_{\text{future op}} - C_{\text{move}})/N_{\text{moves}}$. *In absolute terms (vs never-move baseline), VPM is negative and worsens with λ_m ($-0.8 \rightarrow -84.0$): at high costs, even optimal moves lose vs doing nothing. In relative terms, $VPM_{\text{dynamic}} > VPM_{\text{myopic}}$ at all seven λ_m levels, with the gap widening monotonically (+0.1 $\rightarrow$ +5.5).*

The combined interpretation: *when not to reconfigure* at high λ_m does not mean “never reconfigure”; it means **reconfigure selectively**—only when expected future value exceeds the move cost. The dynamic policy’s advantage grows with stakes precisely because it is more selective per move.

10 Transition Cost: An Inadequate Surrogate

10.1 Hidden Reconfiguration

Finding 10 (Switch-Count Failure). *Across 5 cost conditions $\times$ 6 seeds (36 transitions each): False Switch rate (expert name changed, layout ≤ 2 moves different) = 0%. **Hidden Reconfiguration rate** (expert name unchanged, layout ≥ 20 moves different) = 17–69%. Under switch-only penalties ($\lambda_s = 20$), the rate rises to 69% for the dynamic policy: policies learn to “change content without changing name” to avoid the indicator penalty.*

10.2 Metric Properties

Proposition 1 (Valid Metric Space). *Both $d(L, L') = |\{s : L(s) \neq L'(s)\}|$ (Hamming) and $d(L, L') = \sum_s \|L(s) - L'(s)\|_2$ (Euclidean sum) satisfy symmetry and the triangle inequality on layout pairs, constituting a valid metric space for the MTS framework.*

However, four warehouse-specific phenomena violate clean MTS assumptions: (i) asymmetric open/close pick-face costs; (ii) capacity-infeasible transitions (hard constraint, not merely costly); (iii) batch effects (k simultaneous moves share mobilization cost); (iv) swap chains (exchanging two SKUs’ locations requires a temporary third position). We detected 663 swap pairs across expert layout pairs; each swap’s true labor cost exceeds its Hamming count by ≥ 1 move.

These violations motivate $C_{\text{trans}} = d(L_t, L_{t+1})$ as a *generalized switching cost* rather than a pure metric.

11 Real Warehouse Validation

11.1 WEPA-Natural

40 top-retrieval SKUs, 12,000 real retrieval orders, real layout geometry (1,074 storage cells at 1.4m grid):

Finding 11 (No Natural Trap). *Gap = 0.00%; the myopic trajectory is identical to the BFIP trajectory. Winner diversity exists (E1/E6/E7 alternate by period), but the greedy per-period choice is already globally optimal.*

11.2 WEPA-Stress

Injecting a demand transition (628 surge orders on bottom-half SKUs, $\sim 30\%$ volume increase in one period) followed by a $\times 20$ move-cost shock:

Finding 12 (Moderate Stress Still Insufficient). *Gap remains 0.00%. The surge changes period winners but does not create an inter-temporal trap.*

11.3 External Validity Boundary

Finding 13 (Boundary). *Inter-temporal reconfiguration traps require demand regime changes of a magnitude not present in steady-state warehouse operations. This bounds trap-aware expert routing to promotion seasons, seasonal transitions, and product-line changes—precisely the non-stationary scenarios DWERP targets.*

This is consistent with practical slotting guidelines (e.g., “re-slot only when payback is clear”): the everyday answer to “when not to reconfigure” is *don’t bother*.

12 Discussion

12.1 When Sequential Routing Matters

- **Demand regime change:** promotion ($\times 3\text{--}10$), velocity reversal, affinity shift—sufficient magnitude to change the optimal layout
- **Intermediate reconfiguration cost:** $\lambda_m \sim 10$ (peak value); near-zero (free moves $\rightarrow$ plan per-period) or extreme (nobody moves $\rightarrow$ stickiness is trivially optimal)
- **Intermediate lead time:** $\Delta t \sim 1$ period between demand transition and cost shock

12.2 When It Doesn't

- Steady-state operations (WEPA-Natural: gap = 0)
- Near-zero move costs (planning unnecessary)
- Extremely high move costs (plateau; VPM negative vs baseline)

12.3 The Deployment Implication

The lazy-policy paradox suggests that practical systems should *err toward stickiness* rather than sophistication. A crude forecast model that underestimates optimization opportunities will naturally make fewer moves—and in trap-prone environments, this is protective. This is the warehouse-domain instance of the MTS lazy-vs-aggressive trade-off.

12.4 Design Implications for WMS Vendors

1. Measure $d(L_t, L_{t+1})$, not algorithm switches
2. Cost-aware expert selection: predict $\hat{C}(E_i|S_t)$, not classify winners
3. Option-value hold: when uncertain, deferring a locally beneficial reconfiguration has positive expected value

13 Conclusion

We formalized, demonstrated, and bounded the inter-temporal reconfiguration trap in dynamic warehouse slotting. The core insight—*when not to reconfigure*—is the option value of deferring locally beneficial physical changes under future uncertainty. The deployment paradox (conservatism = protection) suggests that practical systems need not be clairvoyant to perform well; they need to be appropriately reluctant to act.

Three mechanisms, protected throughout this work, constitute the paper's contribution:

1. Different states $\rightarrow$ different experts are optimal (T0 diversity)
2. Current optimum $\neq$ long-run optimum (T1 trap window)
3. As reconfiguration cost grows, *whether to act* becomes the decision variable (T2 sensitivity, VPM)

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